

ALEJANDRO SILVA RODRÍGUEZ

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Málaga, Spain

SUMMARY

Passionate Data Scientist with solid experience in data analysis, machine learning, visualization, and large language models (LLMs). Skilled in Python and R, with the ability to apply advanced AI techniques in real-world settings. I enjoy solving complex and diverse problems through AI-powered solutions, ranging from natural language processing to intelligent automation. Seeking to contribute to innovative projects that create real impact and improve efficiency through data.

EXPERIENCE

- **Accenture**  March 2025 - Present
Full Stack Developer Intern (Java Spring Boot + Angular) Málaga, Spain
 - Developed new features and resolved issues on both the backend and frontend.
 - Improved system stability and usability by working on maintenance and service enhancement tasks.
 - Collaborated with a team of experienced developers, allowing me to acquire best practices and knowledge in software development.

SKILLS

- **Programming Languages & Frameworks:** Python, R, Java, SpringBoot, Angular, Bash
- **Database Systems:** SQLServer, MySQL, MongoDB
- **Data Science & Machine Learning:** Python, R, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Matplotlib
- **Data Engineering:** ETL, Power BI, Data Warehouse, XML, RDF, OWL, SPARQL, MDX
- **Cloud & DevOps Tools:** AWS, Docker, Git, Subversion
- **Soft Skills:** Adaptability, Critical Thinking, Fast Learner, Time Management, Empathy, Initiative.

LANGUAGES

Spanish: Native
English: Advanced (Cambridge C1 Certified)

EDUCATION

- **University of Málaga** Expected Graduation: 2025
Health Engineering (Bioinformatics) Málaga, Spain
 - GPA: 8.33/10

PROJECTS

- **Bachelor Thesis: Optimizing Lightweight Language Models for Clinical Summarization** June 2025
Tools: Python, PyTorch, Hugging Face, Unsloth, FastAPI, ChromaDB, JavaScript 
 - Compared lightweight models (Mistral, Phi, LLaMA) for clinical summarization, fine-tuned LLaMA 3.2, and evaluated using GPT-4o-mini as reference.
 - Implemented multi-query RAG architecture with ChromaDB and integrated it into an interactive web app using FastAPI and JavaScript.
- **Medication Expense Analysis: Data Warehouse for U.S. Hospitals** February 2025
Tools: SQL/OLAP, ETL, MDX, Power BI 
 - Designed a data warehouse and ETL process to analyze medication expenses in ICUs, creating OLAP cubes and MDX queries for advanced analysis.
 - Developed interactive Power BI dashboards to visualize trends and support hospital management decision-making.
- **Breast Cancer Prediction and Analysis** June 2024
Tools: R, Shiny, Quarto, ggplot2  | 
 - Developed and evaluated machine learning algorithms to predict breast cancer survival rates, integrating them into an interactive web app with visualizations to aid interpretation.

COURSES

- **Machine Learning and Big Data for Bioinformatics – University of Granada (100 hours)** April 2023
Verification code: AEFD8A7D8A81B3BF5E0B0EEFF5A82287